Recording

EN 580.133
ASOs as Therapeutics (Part II)

Elaine Zhelan Chen

Jan 14, 2025

Created by Elaine Zhelan Chen

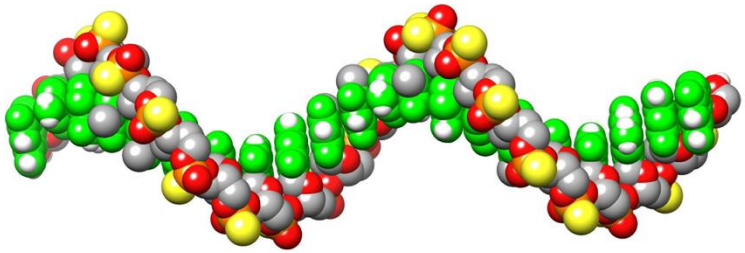
Recap

- Pharmacology terms, PKPD, EC50, IC50, therapeutic index
- siRNA backbone stabilization through chemical modification
  - STC, initial introduction of stabilized siRNA backbone
  - ESC, more 2’OMe substitution, increased siRNA potency
  - ESC+, incorporation of GNA to minimize off-target effects
- ASO mechanisms
  - RNase H
  - Steric block
    - modulation of alternative splicing
    - Translation inhibition
    - Increase translation

ASO drug challenges

- Single-stranded ASOs are ~4–10 kDa (siRNAs are ~14 kDa)
- 13 ASO drugs approved

- Do not readily pass through the plasma membrane
- Prone to nuclease degradation
- Fast clearance by the kidney
- Challenging delivery
    - So far, approved ASO drugs are largely locally delivered or targeted towards the liver



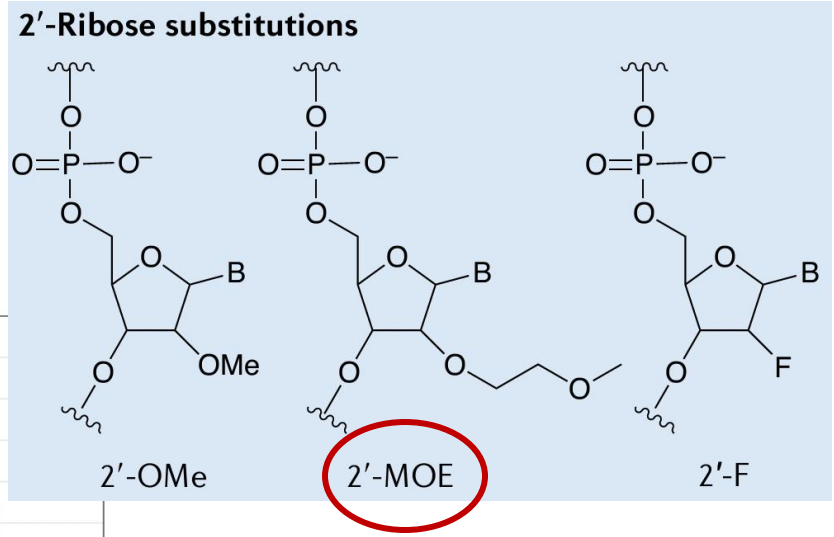
Chemical modification

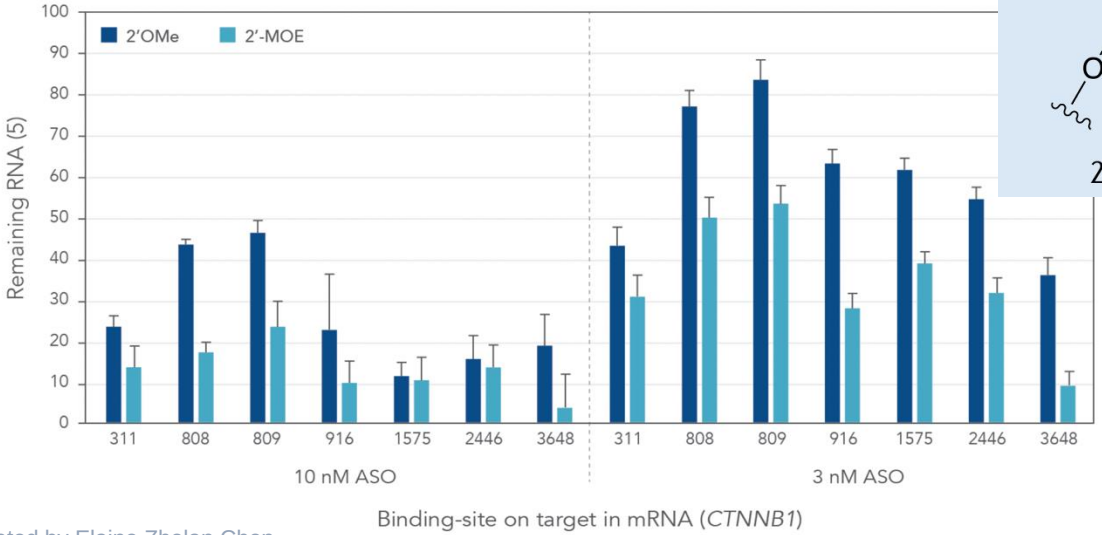
- Extensive chemical modification allows ASO delivery without the need for an additional vehicle
- Most approved oligonucleotide therapies are ‘naked’ ASOs

- Key chemical modifications in clinically approved ASOs
  - Phosphorothioate (PS)
  - 2’-O-MOE, 2’-F, 2’-OMe
  - LNAs
  - PMO

Chemical modification - 2’-MOE

- 2’ O-methoxyethyl (2’-MOE)
- Increased nuclease resistance
- Improved stability in plasma
- Increased tissue half-lives
- Prolonged drug effects
- Better affinity to the target

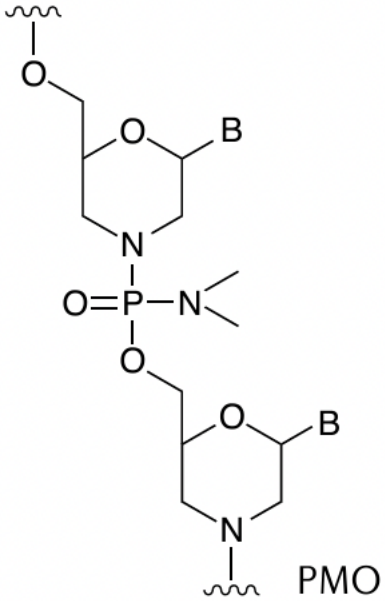




Chemical modification – PMO

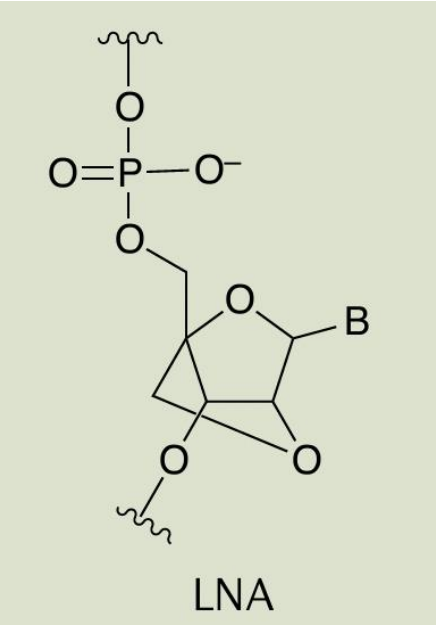
- PMO (phosphorodiamidate morpholino oligonucleotide)
- The five-membered ribose ring is replaced by a six-membered morpholine ring
- Charge-neutral nucleic acid chemistry

- Significantly enhanced metabolic stability
  - Additional resistance against proteases and esterases due to loss of the carbonyl group
- Similar binding affinity to RNA as DNA
- Low protein binding due to the uncharged backbone
  - Downside: fast clearance
- Do not support RNase H activity
  - Used in steric block ASOs for splice correction



Chemical modification - LNA

- Locked nucleic acids (LNAs)
  - An extra bridge connecting the 2' oxygen and 4' carbon
- Enhance nuclease stability
- Better affinity of the ASO for target RNA
  - Incorporated into the flanking regions of gapmers to improve target binding
  - Better potency but has shown more toxicity
- Not compatible with RNase H activity
  - Excluded from the DNA gap region
  - Used in steric block ASOs



Approved ASO drugs

Landscape of approved drugs

- RNaseH1-mediated mRNA cleavage
- Steric block ASO by modulation of alternative splicing

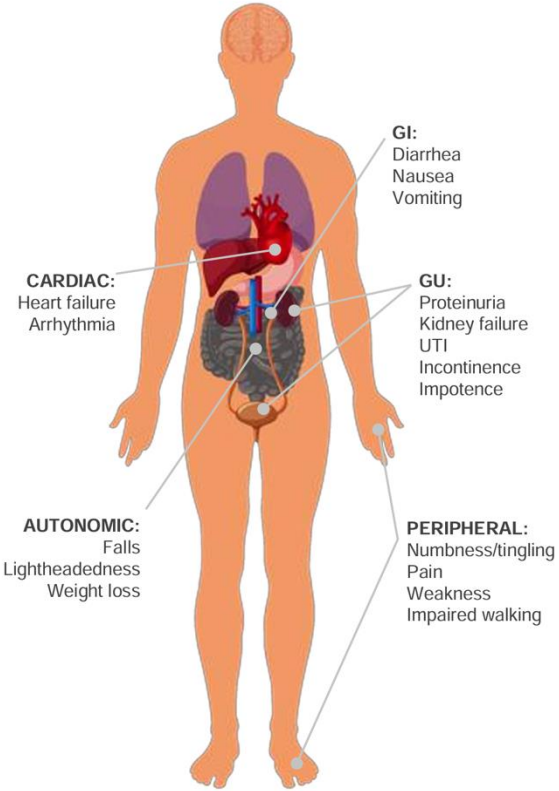
- 4 clinically approved drugs
  - Case #1 hATTR
  - Case #2 DMD
  - Case #3 SMA
  - Case #4 Batten disease

Case #1 hATTR

Hereditary transthyretin-mediated amyloidosis (hATTR)

- TTR transports the thyroid hormone thyroxine and vitamin A
- Patisiran lowers the TTR level by 91%

- TTR has neuroprotective functions
- TTR KO mice are viable, although they exhibit alterations in behavior and memory
- Whether partial or total reduction of TTR function in humans will be associated with deleterious effects is not clear

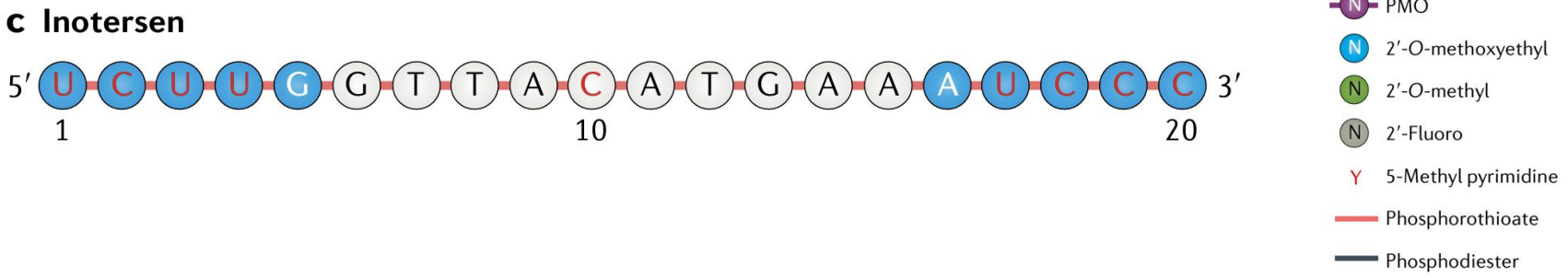


Inotersen

- TEGSEDI (inotersen)
- Approved in 2018 for treating hATTR
- Developed by Ionis Pharmaceuticals

- A PS-modified 5-10-5 gapmer with MOE flanks and a central DNA gap
  - MOE increases affinity to the target

- Patisiran has shown better effects on neuropathy (a symptom of hATTR) than Inotersen



GalNAc-conjugated ASO

- Wainua (eplontersen)
- For treating polyneuropathy by hATTR
- Developed by AstraZeneca and Ionis Pharmaceuticals
- Approved in 2023

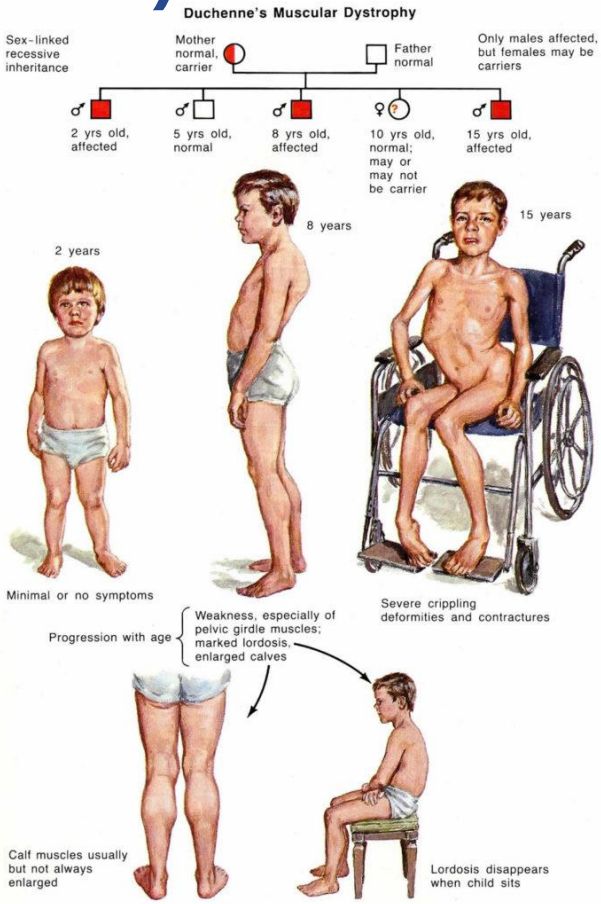
- Self-administered via an auto-injector, subcutaneous
- GalNAc-conjugated eplontersen is used at a lower dose than the unconjugated inotersen

Case #2 DMD and BMD

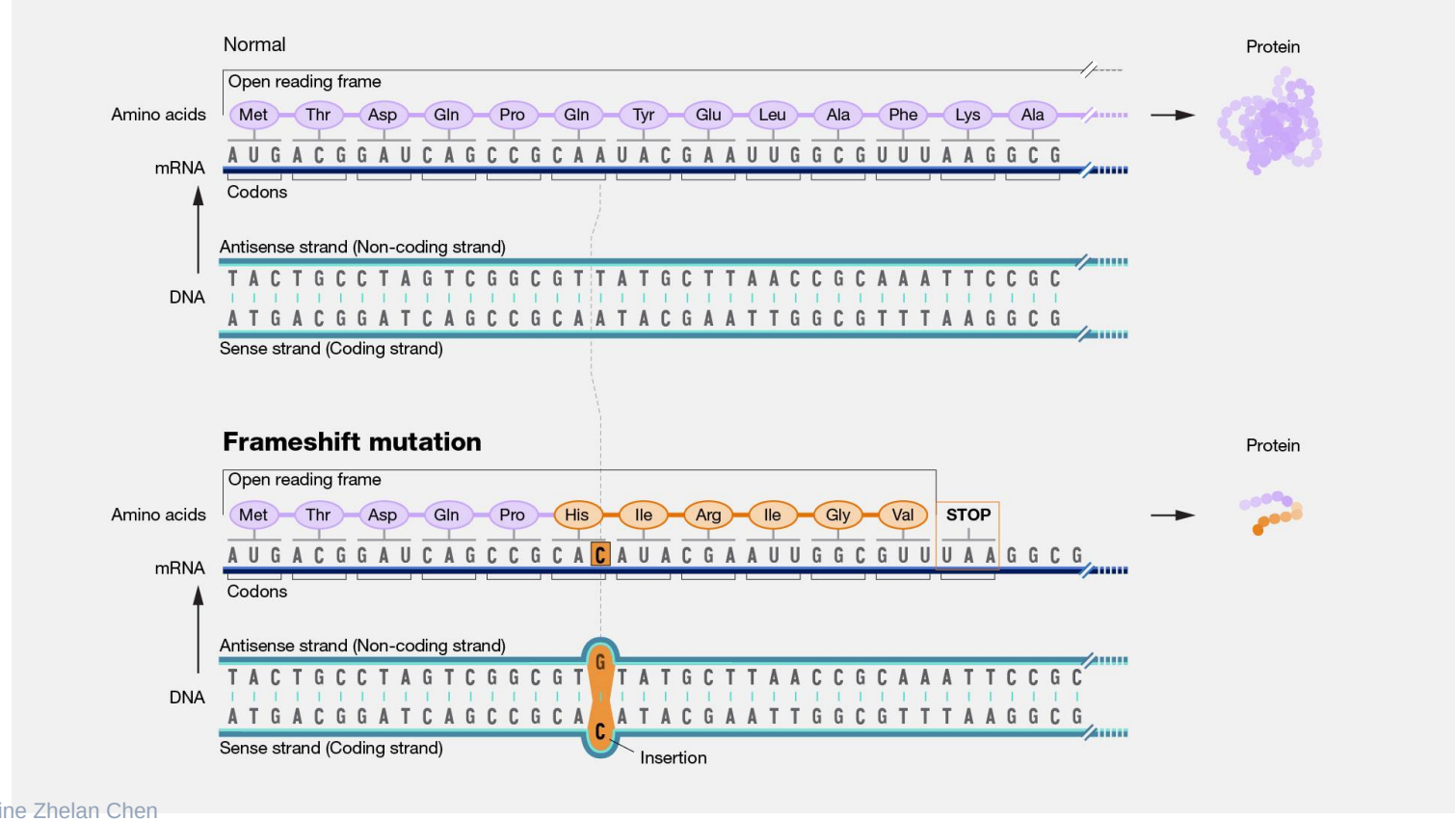
Duchenne muscular dystrophy (DMD)

- DMD is a fatal muscle degenerative disorder
- It is caused by mutations in the DMD gene, which encodes dystrophin
- The partial gene deletion causes a reading frame shift resulting in an early stop codon, leading to a lack of functional dystrophin protein

- Progressive muscle degeneration and weakness
- X-linked, occurs more often in males
  - 1 in 3,500–5,000 males born worldwide
  - Lose the ability to walk by 12 years of age
  - Death often occurs within the 20s, usually due to respiratory or cardiac complications



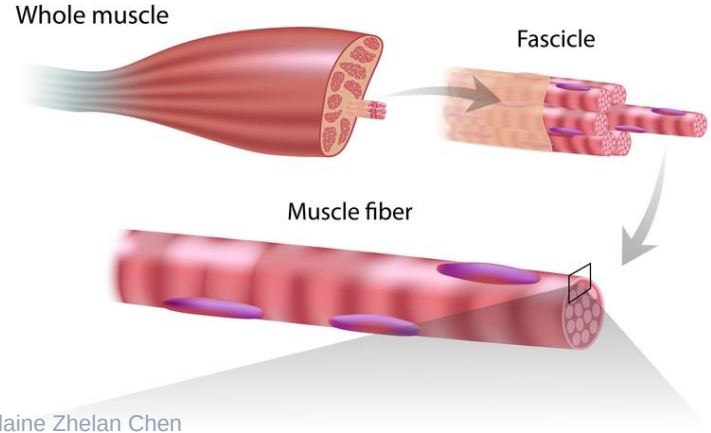
Frameshift mutation

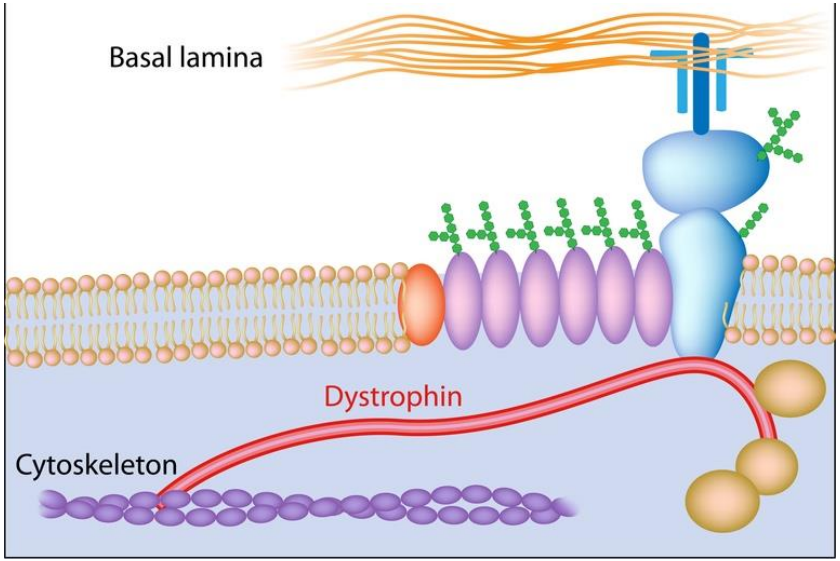


Created by Elaine Zhelan Chen

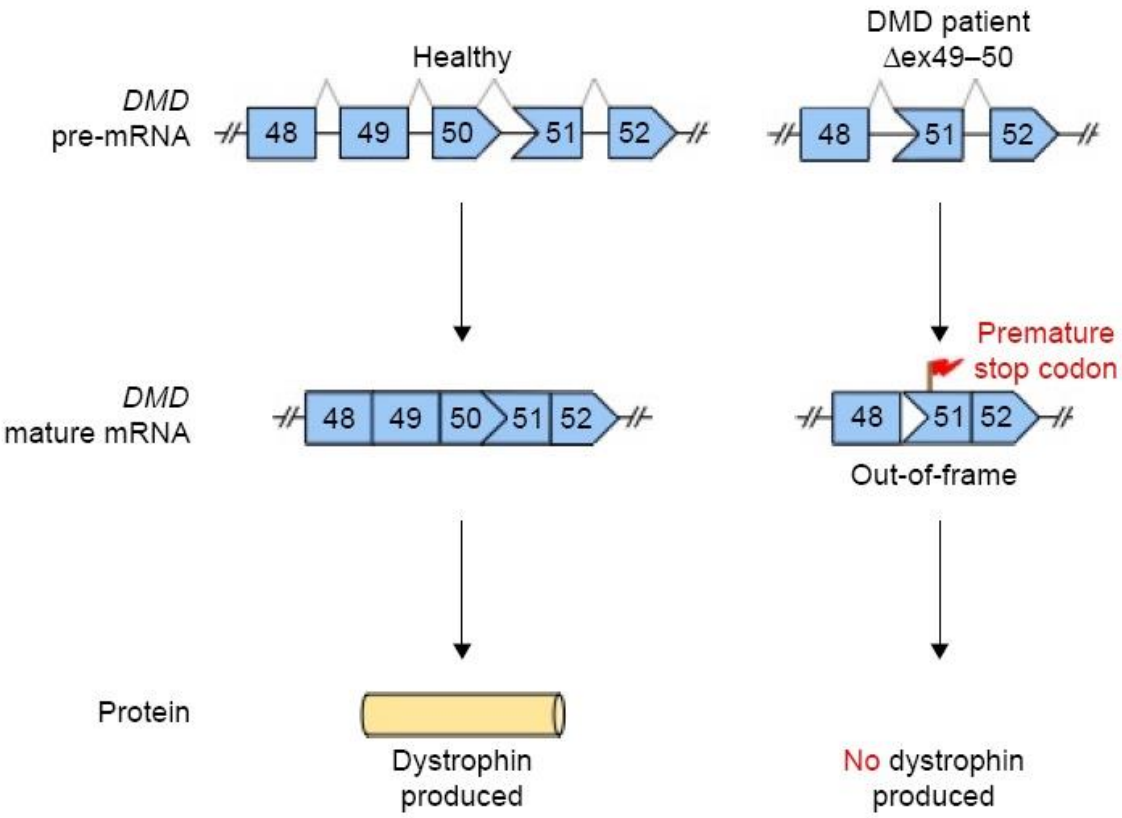
Dystrophin

- The largest known human gene
  - Containing 79 exons and spanning > 2,200 kb, roughly 0.1% of the whole genome
- Dystrophin connects intercellular cytoskeleton to extracellular matrix and provides biomechanical support to the skeletal or cardiac plasma membrane during contraction
- Loss of dystrophin predisposes muscle fibers to mechanical damage, leading to muscle degeneration.



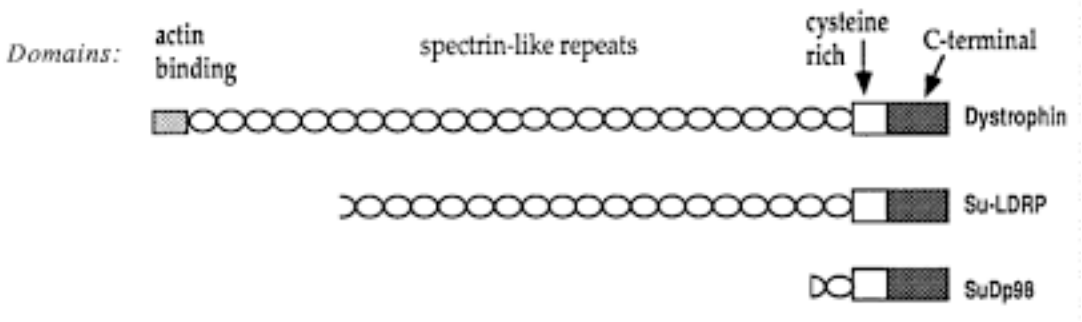


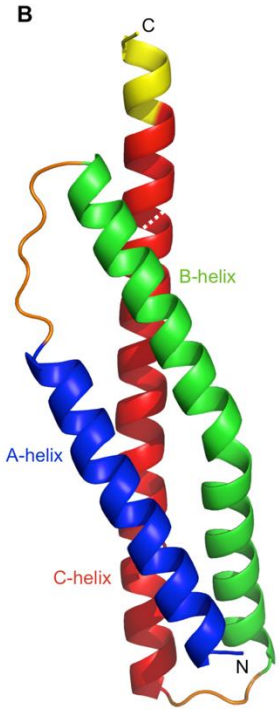
DMD-causing mutation



Dystrophin spectrin repeats

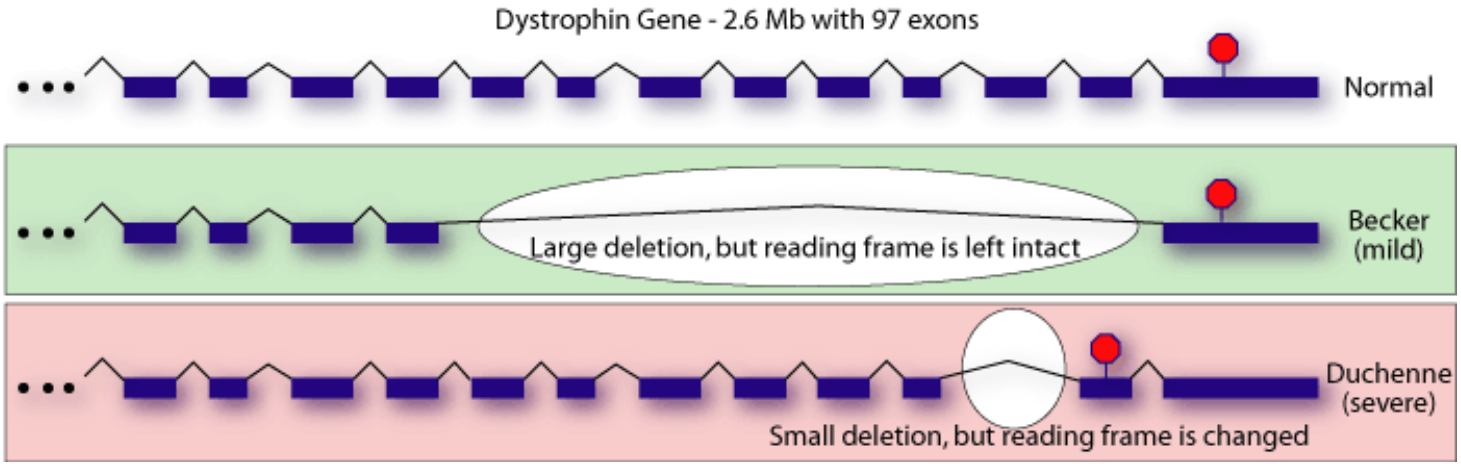
- The central rod of dystrophin is composed of 24 spectrin-like repeat
- Each repeat unit is ~110 aa in size and forms a triple α-helical bundles





Becker's Muscular Dystrophy (BMD)

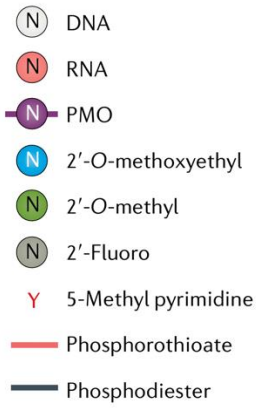
- BMD is a rare, inherited condition that causes muscle weakness which worsen over time
- Like DMD, BMD is also caused by mutations in the DMD gene, which encodes dystrophin
- Unlike DMD, BMD is caused by in-frame mutations in the DMD gene
  - Results in a shortened but partially functional dystrophin protein
  - BMD patients have less severe symptoms

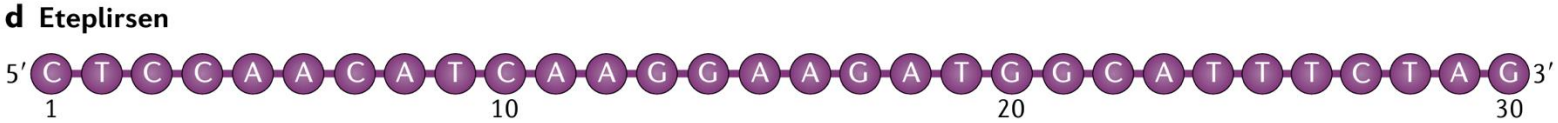


Eteplirsen

- EXONDYS 51 (eteplirsen)
- Developed by Sarepta
- Approved in 2016

- A PMO 30mer, containing PMO-based uncharged backbone
- Functions through alternative splicing

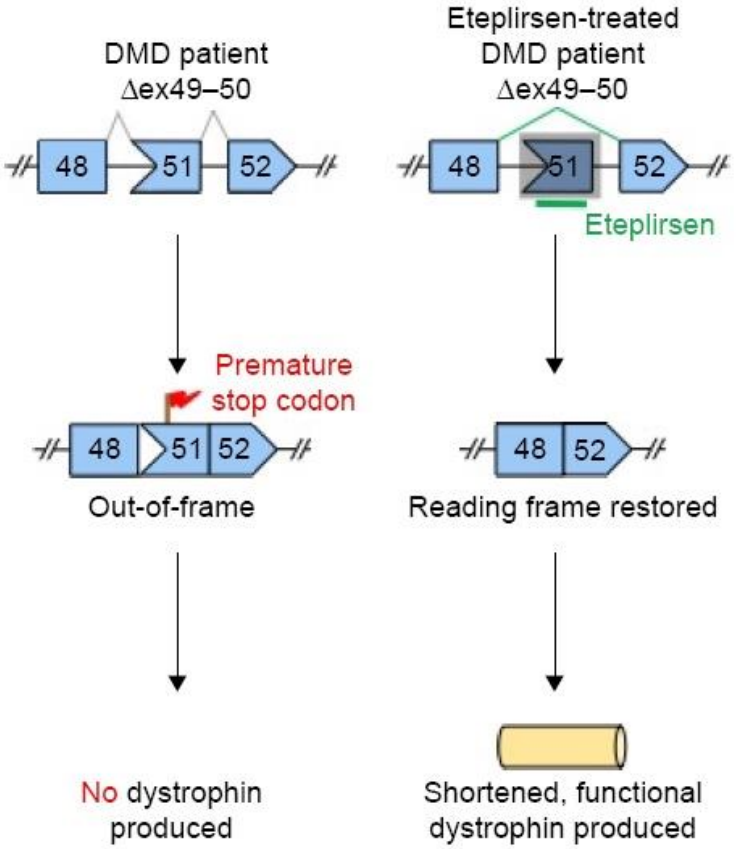




Eteplirsen mode of action

- Eteplirsen binds to exon 51 of DMD
- Restores the reading frame of the dystrophin mRNA
- Allows translation of a short, but functional dystrophin

- Patients receiving Eteplirsen showed an increase in dystrophin levels
- Eteplirsen received FDA approval after considering the lack of available treatment options for DMD



Eteplirsen therapeutic effects

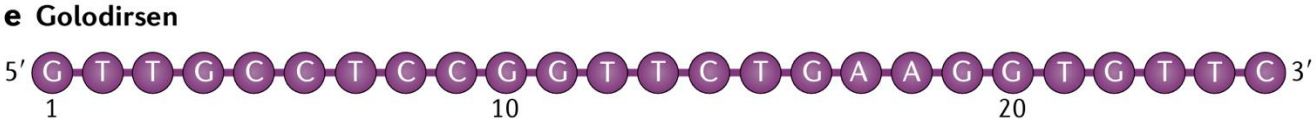
- Significant reductions in rates of hospital encounters (31%), emergency room visits (31%), need for pulmonary management (33%), cardiac management (21%), tracheostomy (86%), and assisted ventilation (39%)

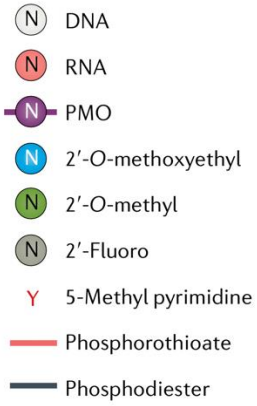
- Ongoing post-approval clinical studies to validate the efficacy of Eteplirsen in decelerating the progression of DMD

Golodirsen

- VYONDYS 53 (golodirsen)
- Developed by Sarepta
- Approved in 2019

- A PMO 25mer, containing PMO-based uncharged backbone
- Targets exon 53 of the dystrophin transcript (about 8% of the DMD community)
- Functions through alternative splicing



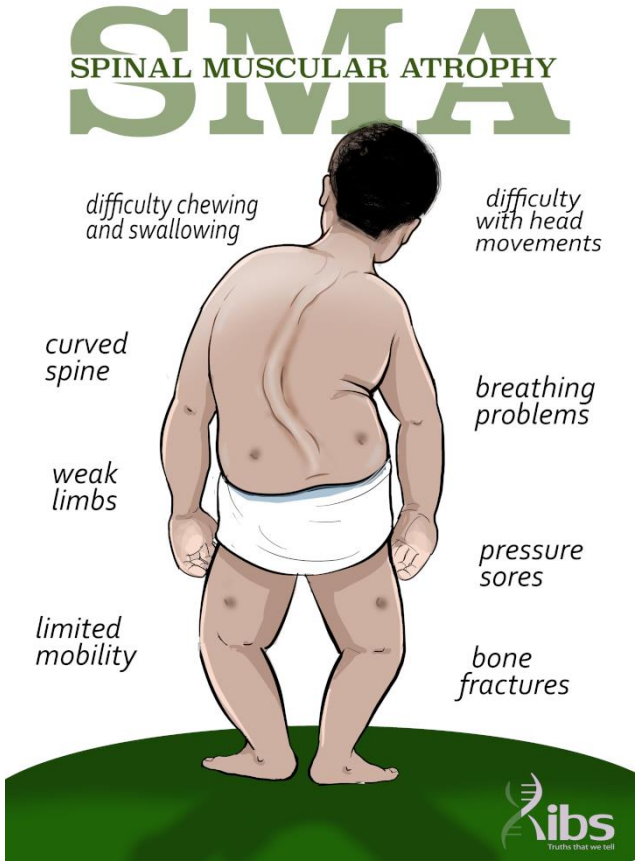


Case #3 SMA

Spinal Muscular Atrophy (SMA)

- SMA is the second most common fatal autosomal recessive disorder
- Leading cause of death in early childhood by a monogenic disease
- incidence of ~1 in 6,000 to 10,000 births
- SMA is caused by mutations in both copies of the SMN1 (survival motor neuron) gene located on chromosome 5

- SMA affects the motor neurons
- The muscles weaken and shrink from inactivity since they cannot respond to signals from the nerves
- Conditions gets worse as over time
- Most cases do not survive beyond 2 years of age

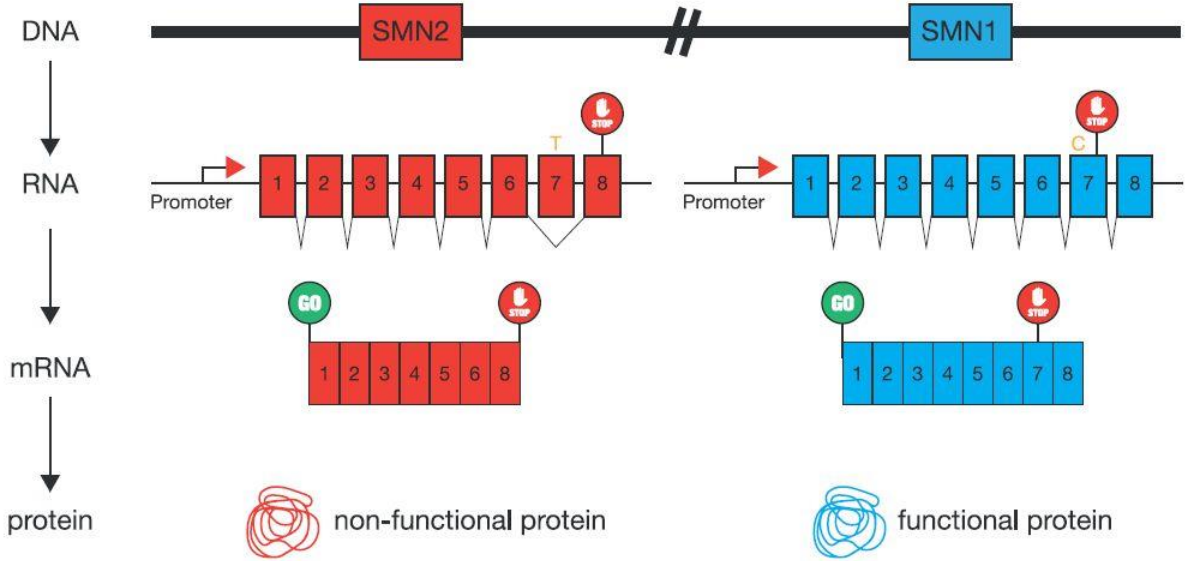


SMN1 vs SMN2

- In healthy individuals, SMN1 encodes for the survival motor neuron (SMN) protein
- SMN2 differs from SMN1 by one NTP at the 840th position on the gene, T for SMN2 and C for SMN1
    - This leads to different splicing patterns of SMN1 and SMN2

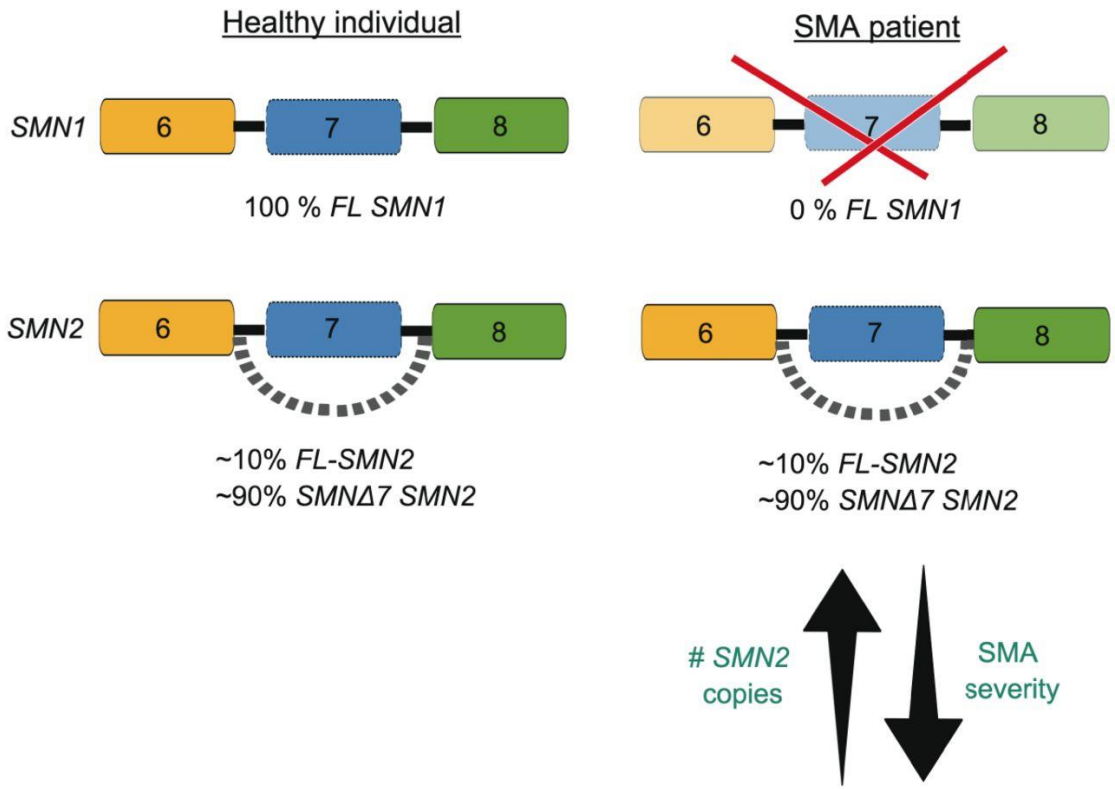
SMN1 vs SMN2

- C-> T results in the exclusion of exon 7 from ~85% SMN2 during splicing
- This results in the production of a truncated SMN protein
  - Unstable and rapidly degraded
- The remaining 15% of SMN2 mRNA include exon 7 and produces full-length functional SMN



Nusinersen mode of action

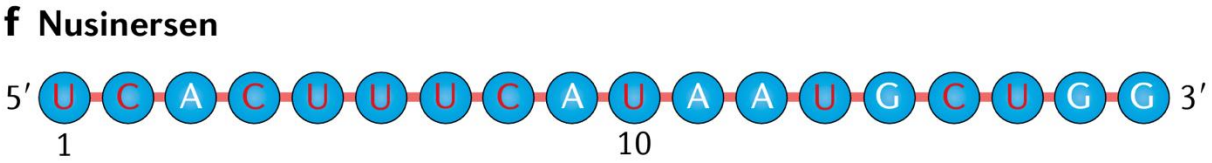
- Nusinersen hybridizes with SMN2 pre-mRNA
- Prevents the recruitment of splicing repressor
- Favors the inclusion of exon 7 in the SMN2 mRNA
- Leads to increase in the production of functional SMN protein form

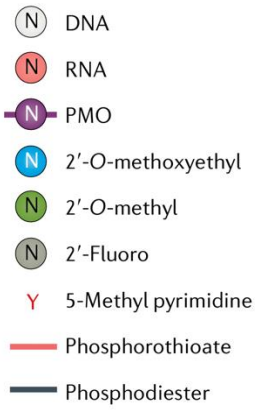


Nusinersen

- Spinraza (Nusinersen)
- Developed by Biogen
- Approved in 2016

- All-PS/MOE-modified RNA 18mer
- For treating SMA by modulating alternate splicing of the SMN2 gene

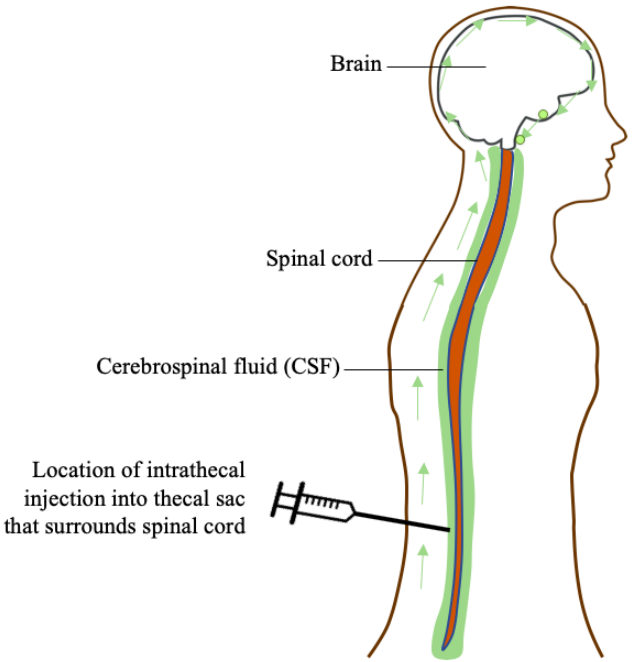




Nusinersen therapeutic effects

- Proved to be efficacious with early treatment initiation
  - Able to halt disease progression
  - Able to develop further and gain developmental milestones not seen before

- A dose of 12 mg Nusinersen is given intrathecally on the first, second, fourth, and ninth week, followed by maintenance therapy after every four-month interval



Case #4 Batten disease

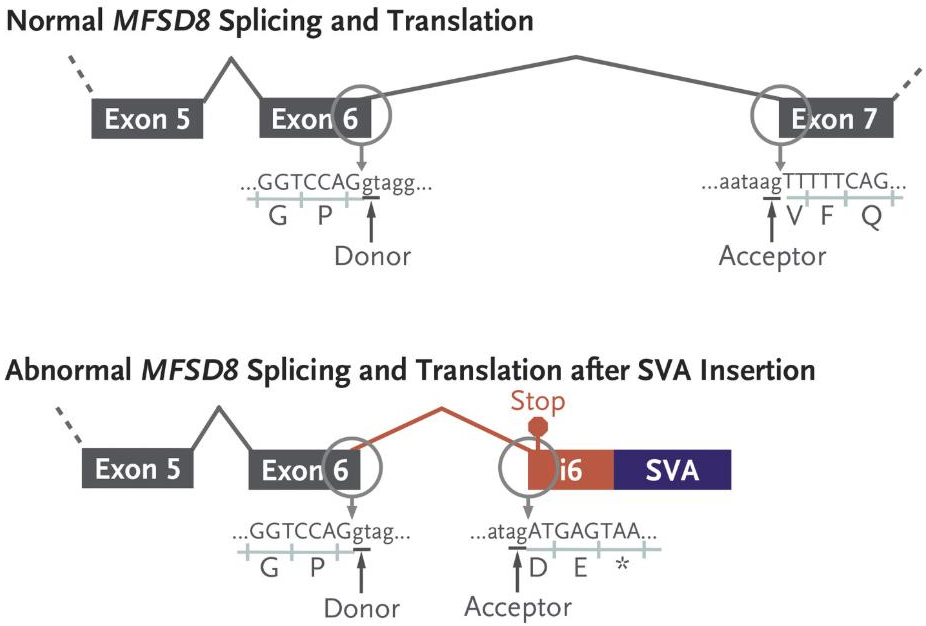
A tailored ASO drug

- To treat a six-year-old child suffering from neuronal ceroid lipofuscinosis 7 (CLN7)
- A rare and fatal neurodegenerative condition, also known as Batten disease
- Leads to loss of vision, seizures, difficulty in speaking and swallowing

- Boston Children's Hospital
- Created by Dr. Timothy Yu and his team
- In 2017

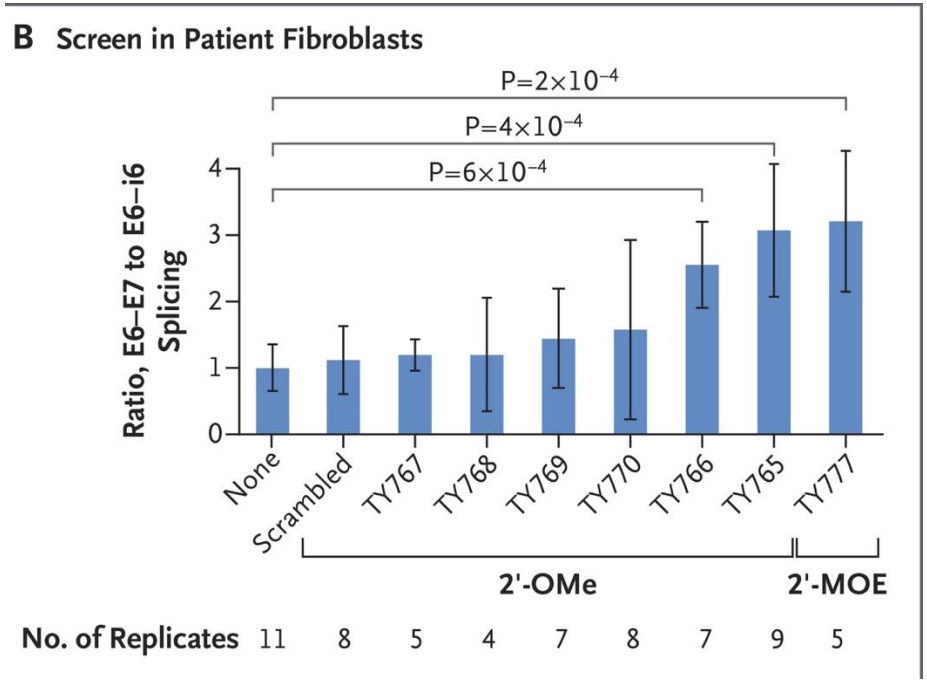
Genetic cause of the disease

- Retrotransposon insertion into the MFSD8 gene, which is mistaken as an exon and led to abnormal splicing
- The team decided to design an ASO drug to correct the splicing



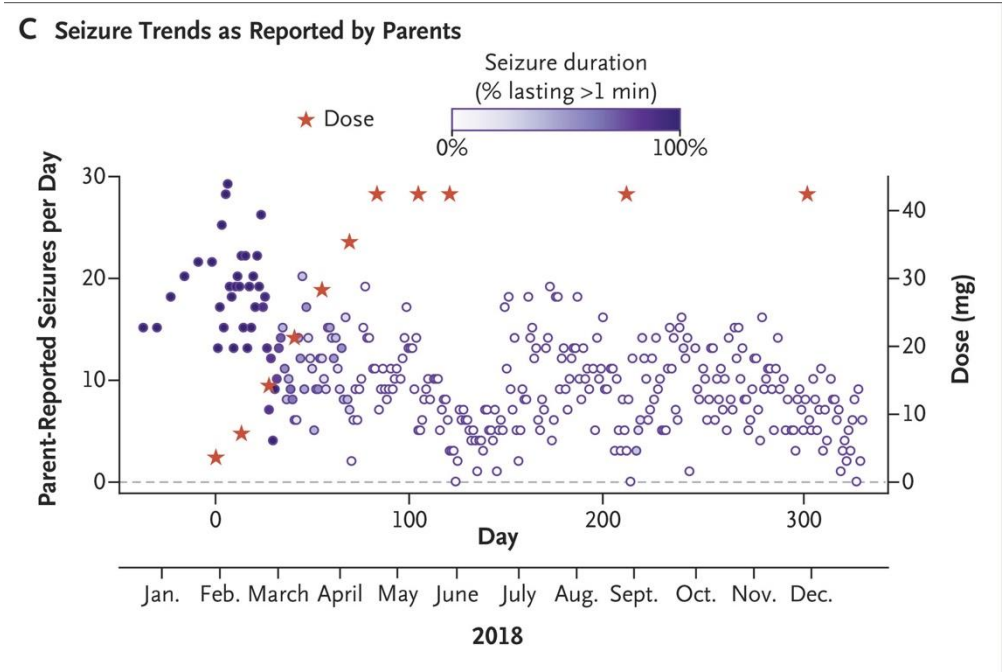
Milasen

- Milasen is a 22-nucleotide ASO with the same backbone and sugar chemistry modifications (PS and 2′-MOE) as nusinersen



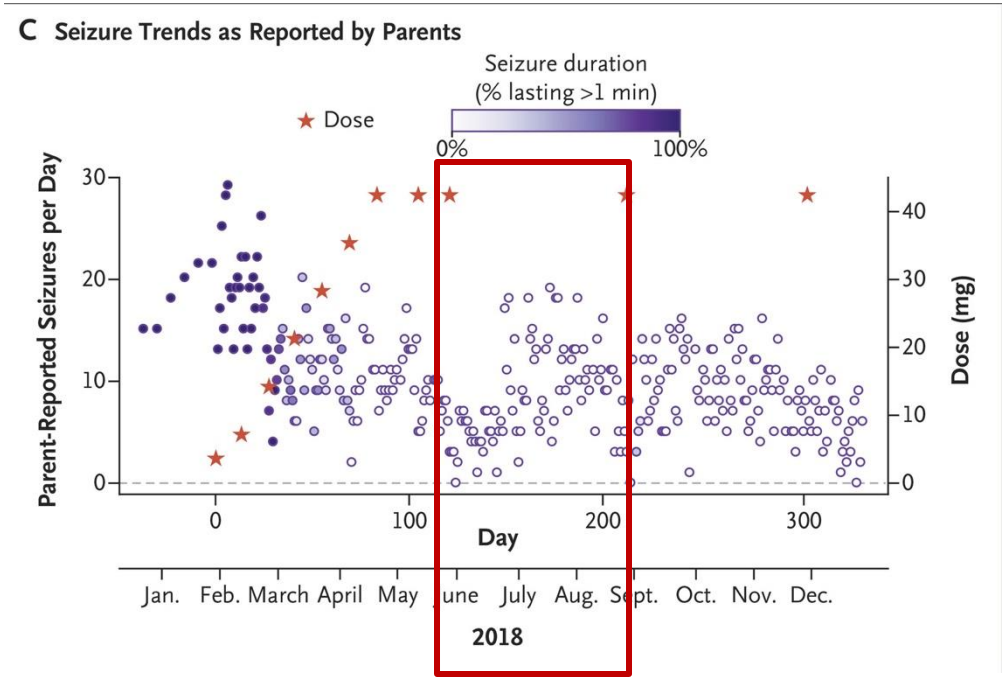
Milasen

- Mila received the first of a series of nine escalating, biweekly doses of milasen given under general anesthesia
- In August 2018, she received her first maintenance treatment, given every three months



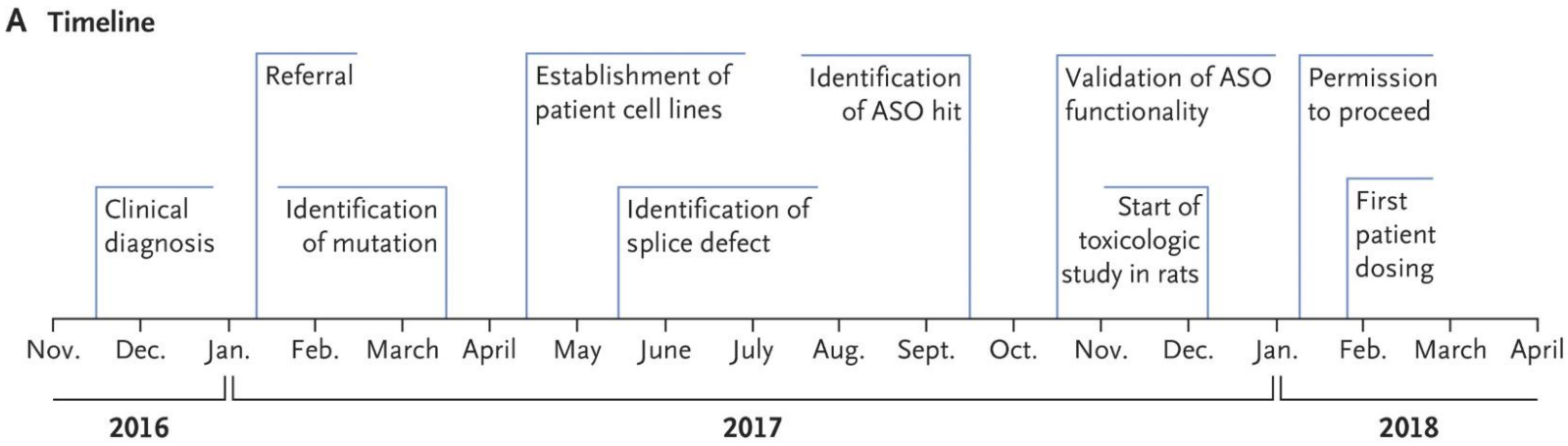
Milasen therapeutic effects

- Mila received the first of a series of nine escalating, biweekly doses of milasen given under general anesthesia till June 2018
- In August 2018, she received her first maintenance treatment, given every three months
- An increased frequency to every two months starting in March 2019
- Less frequent seizures and later barely noticeable
- Improved quality of life for the patient



Milasen development timeline

- Showing the promise of RNA-based personalized medicine



Reminders

- **Problem Set #3 Due Tomorrow, Jan 15th, 5PM EST**

- **Final presentation**
  - **Submit a copy of slides by Tomorrow, Jan 15th, 10PM EST**
  - **Submit teammate evaluation by Thursday, Jan 16th, 5PM EST**

Attendance